

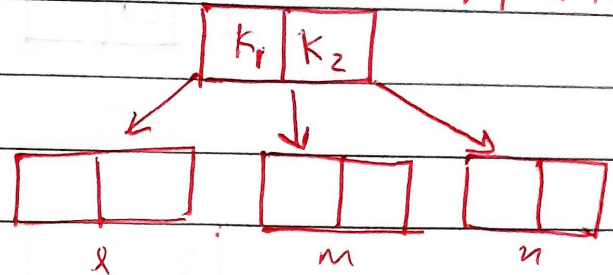
2-3 Trees (Search trees)

Multiway search Tree (M-way)

Degree 3

B-Tree

All leaf nodes at same level.



$$k_1 < k_2$$

Every node must

have $\lceil \frac{n}{2} \rceil = \lceil \frac{3}{2} \rceil = 2$ children

$$l < k_1$$

$$k_2 < n$$

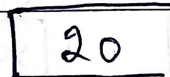
$$k_1 < m < k_2$$

Insertion 2-3 Trees

Keys $\rightarrow$ 20, 30, 40, 50, 60, 10, 15, 70, 80, 90

Note :- 2-3 Trees grow upward. ★

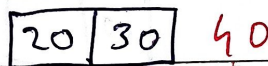
insert 20



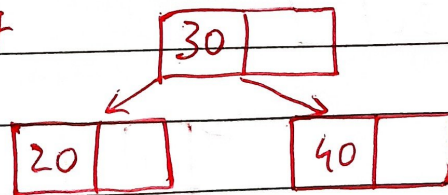
insert 30



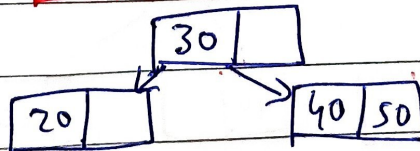
insert 40



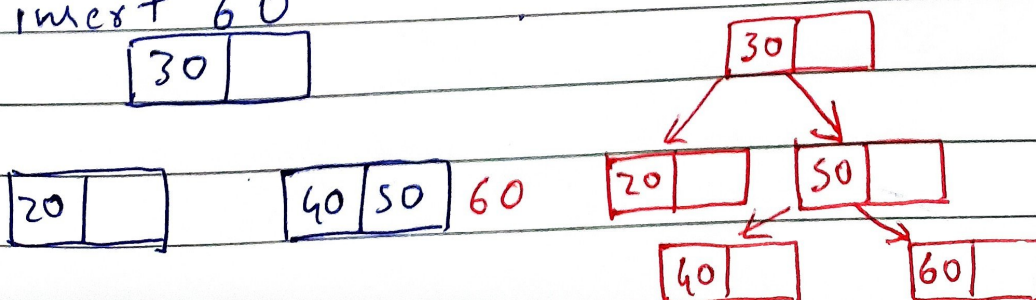
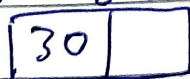
split



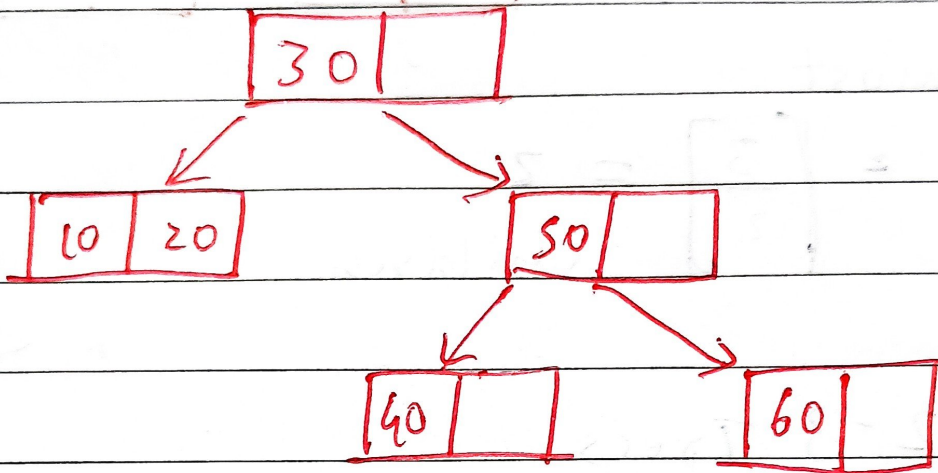
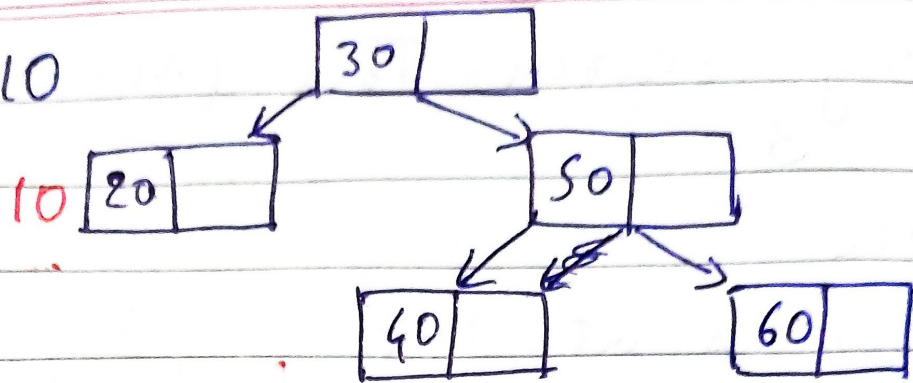
insert 50



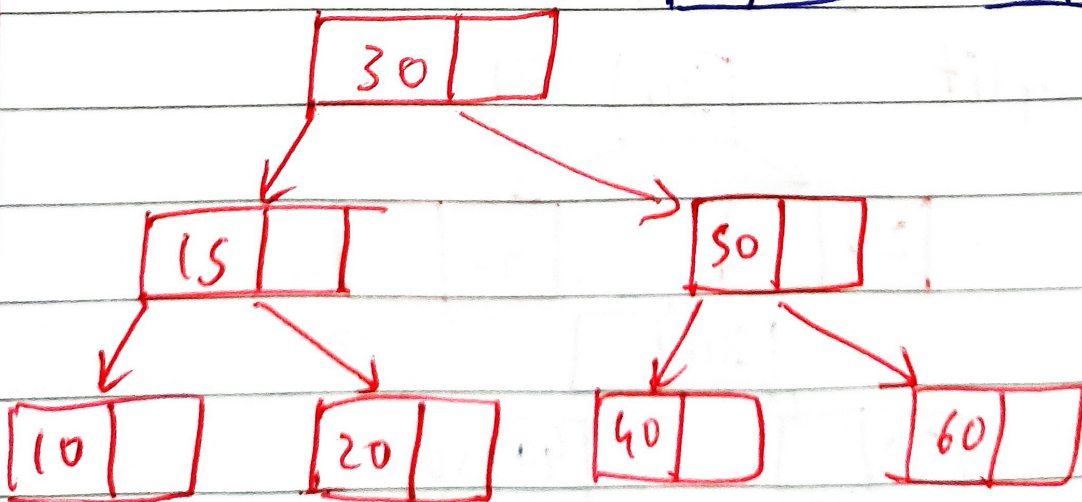
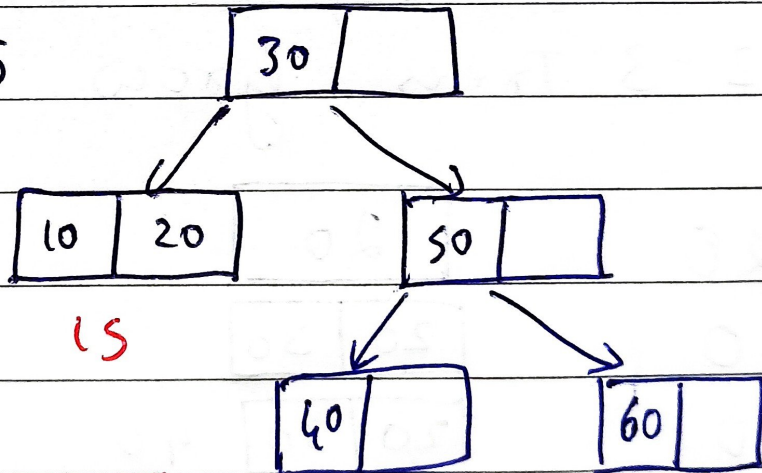
insert 60



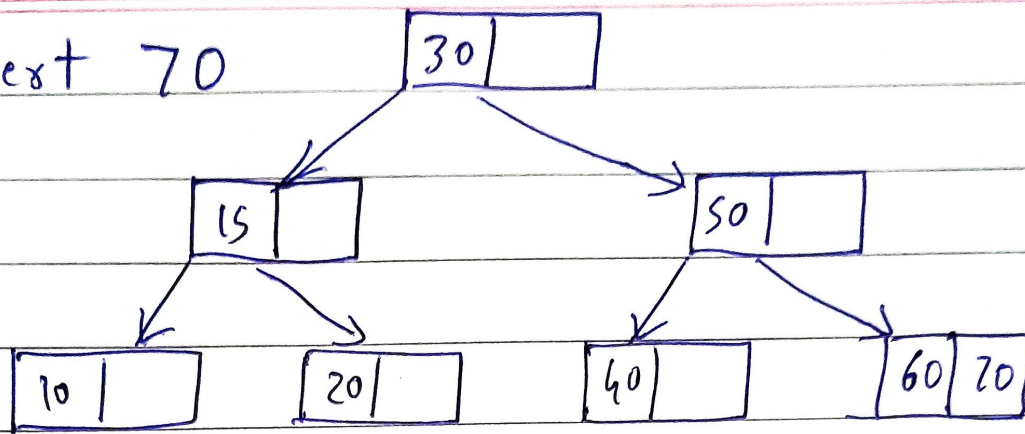
insert 10



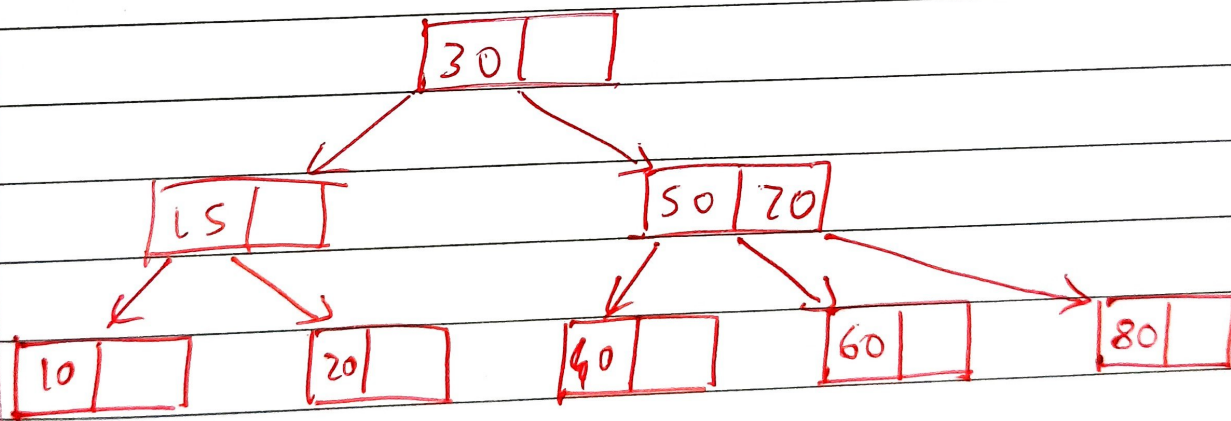
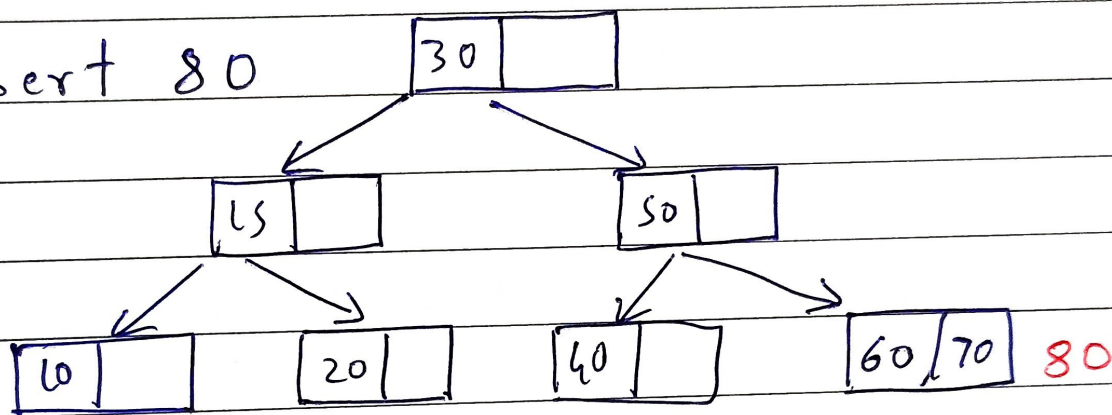
insert 15



insert 70



insert 80



insert 90

